Chebyshev Polynomials

A familiar collection of multiple-angle formulas is

$$\begin{aligned}
\cos(0\theta) &= 1,\\
\cos(\theta) &= \cos\theta,\\
\cos(2\theta) &= 2\cos^2\theta - 1,\\
\cos(3\theta) &= 4\cos^3\theta - 3\cos\theta,\\
\cos(4\theta) &= 8\cos^4\theta - 8\cos^2\theta + 1,\\
\cos(5\theta) &= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta,\\
\cos(6\theta) &= 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1\\
\cos(7\theta) &= 64\cos^7\theta - 112\cos^5\theta + 56\cos^3\theta - 7\cos\theta,\\
\cos(8\theta) &= 128\cos^8\theta - 256\cos^6\theta + 160\cos^4\theta - 32\cos^2\theta + 1.
\end{aligned}$$

These formulas suggest a striking observation: for every nonnegative integer n, $\cos(n\theta)$ can be written as a polynomial in $\cos\theta$.

Let

$$x = \cos\theta.$$

We are therefore led to the polynomials

$$\begin{aligned}
T_0(x) &= 1,\\
T_1(x) &= x,\\
T_2(x) &= 2x^2 - 1,\\
T_3(x) &= 4x^3 - 3x,\\
T_4(x) &= 8x^4 - 8x^2 + 1,\\
T_5(x) &= 16x^5 - 20x^3 + 5x,\\
T_6(x) &= 32x^6 - 48x^4 + 18x^2 - 1,\\
T_7(x) &= 64x^7 - 112x^5 + 56x^3 - 7x,\\
T_8(x) &= 128x^8 - 256x^6 + 160x^4 - 32x^2 + 1,
\end{aligned}$$

which satisfy

$$\boxed{T_n(\cos\theta) = \cos(n\theta), \qquad \theta \in \mathbb{R}.}$$

These are the *Chebyshev polynomials of the first kind*.

The identity

$$T_n(\cos\theta) = \cos(n\theta)$$

holds for every $\theta \in \mathbb{R}$. However, by the periodicity and symmetry of the cosine function, it is sufficient to consider

$$0 \le \theta \le \pi.$$

On this interval, the map $x = \cos\theta$ is one-to-one and maps $[0, \pi]$ onto $[-1, 1]$. This leads to the following definition. We define $T_n(x)$, the *Chebyshev polynomial of degree (n)*, by

$$\boxed{T_n(x) = \cos\big(n \arccos(x)\big), \qquad x \in [-1, 1].}$$

As we have seen above, this expression is in fact a polynomial in x, of degree n. Thus, although the formula involving $\arccos(x)$ is naturally restricted to $x \in [-1, 1]$, the polynomial T_n itself is defined for all $x \in \mathbb{R}$ (and, more generally, for all $x \in \mathbb{C}$).

The recurrence

The addition formula, $\cos(a + b) = \cos(a)\cos(b) - \sin(a)\sin(b)$, gives

$$\cos((n + 1)\theta) + \cos((n - 1)\theta) = 2\cos\theta\cos(n\theta).$$

Hence

$$\cos((n + 1)\theta) = 2\cos\theta\cos(n\theta) - \cos((n - 1)\theta).$$

Replacing $\cos\theta$ by x gives the recurrence

$$\boxed{T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x),}$$

with

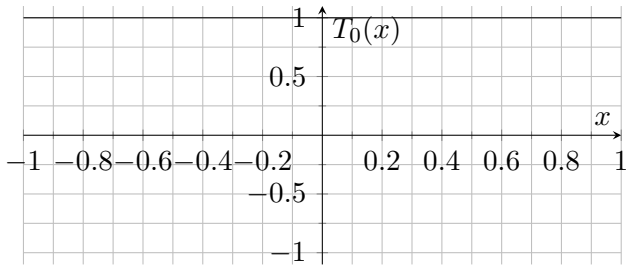
$$T_0(x) = 1, \qquad T_1(x) = x.$$

Thus the entire sequence of Chebyshev polynomials can be generated recursively.

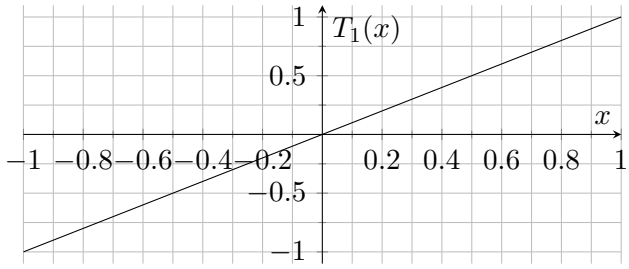
The first nine polynomials

The graphs below show $T_0, \ldots, T_8$ separately. In each case the horizontal axis is $x \in [-1, 1]$, and the same vertical scale is used.

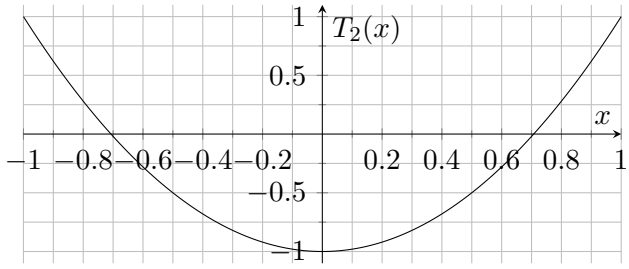
$T_0(x) = 1$



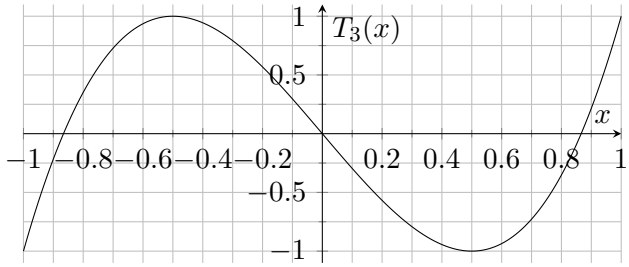
$T_1(x) = x$



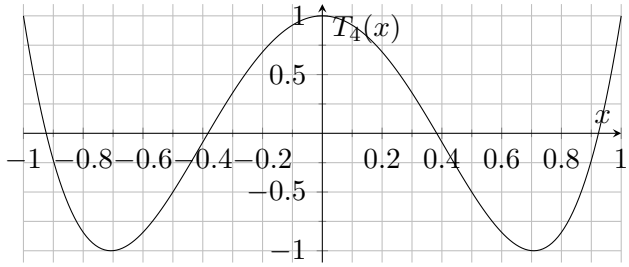
$T_2(x) = 2x^2 - 1$



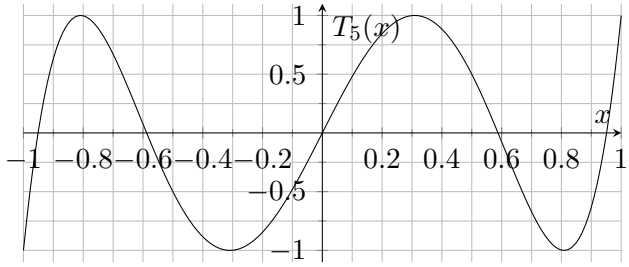
$$T_3(x) = 4x^3 - 3x$$



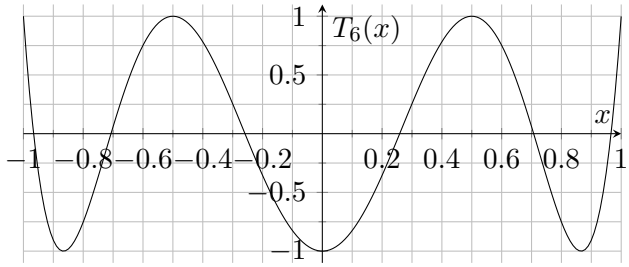
$$T_4(x) = 8x^4 - 8x^2 + 1$$



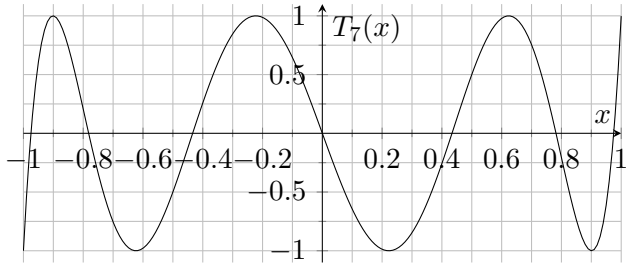
$$T_5(x) = 16x^5 - 20x^3 + 5x$$



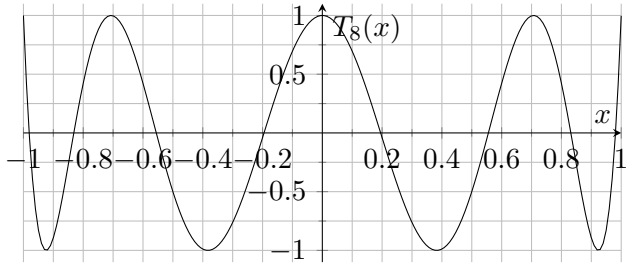
$$T_6(x) = 32x^6 - 48x^4 + 18x^2 - 1$$



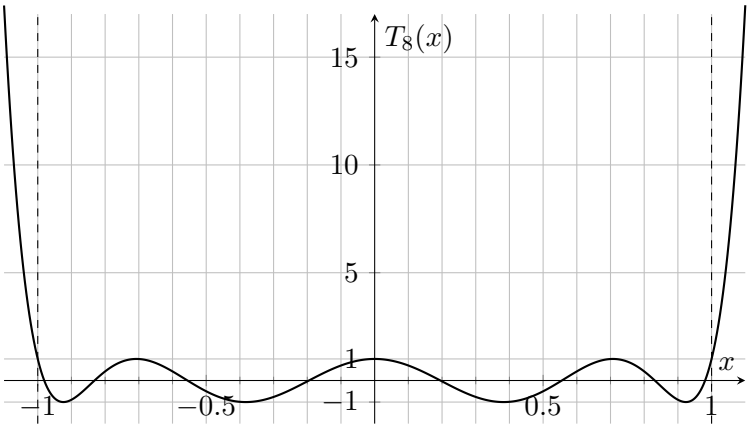
$$T_7(x) = 64x^7 - 112x^5 + 56x^3 - 7x$$



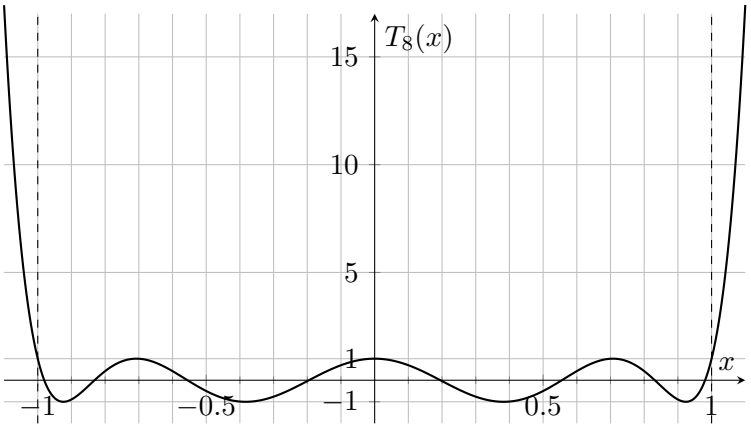
$$T_8(x) = 128x^8 - 256x^6 + 160x^4 - 32x^2 + 1$$

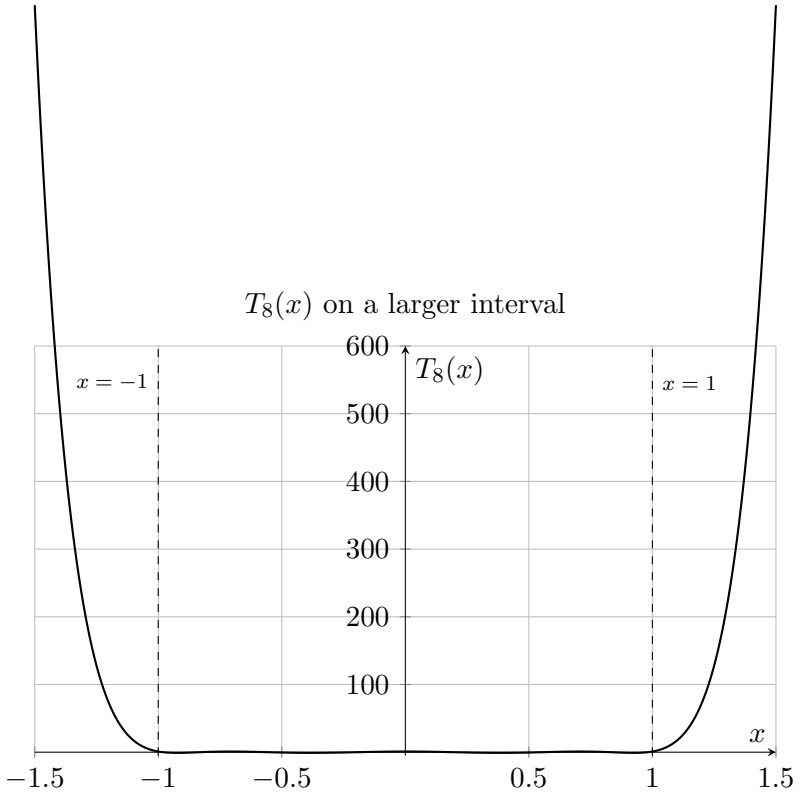


$T_8(x)$ on the interval $[-1.1, 1.1]$



$T_8(x)$ beyond the interval $[-1.1, 1.1]$





One way to visualize a Chebyshev polynomial is to start with the graph

$$y = \cos(n\theta), \qquad 0 \le \theta \le \pi,$$

drawn on a flexible sheet of paper.

Now imagine bending the sheet so that the horizontal coordinate θ is replaced by

$$x = \cos\theta.$$

The vertical coordinate is left unchanged. Thus, the point

$$(\theta, \cos(n\theta))$$

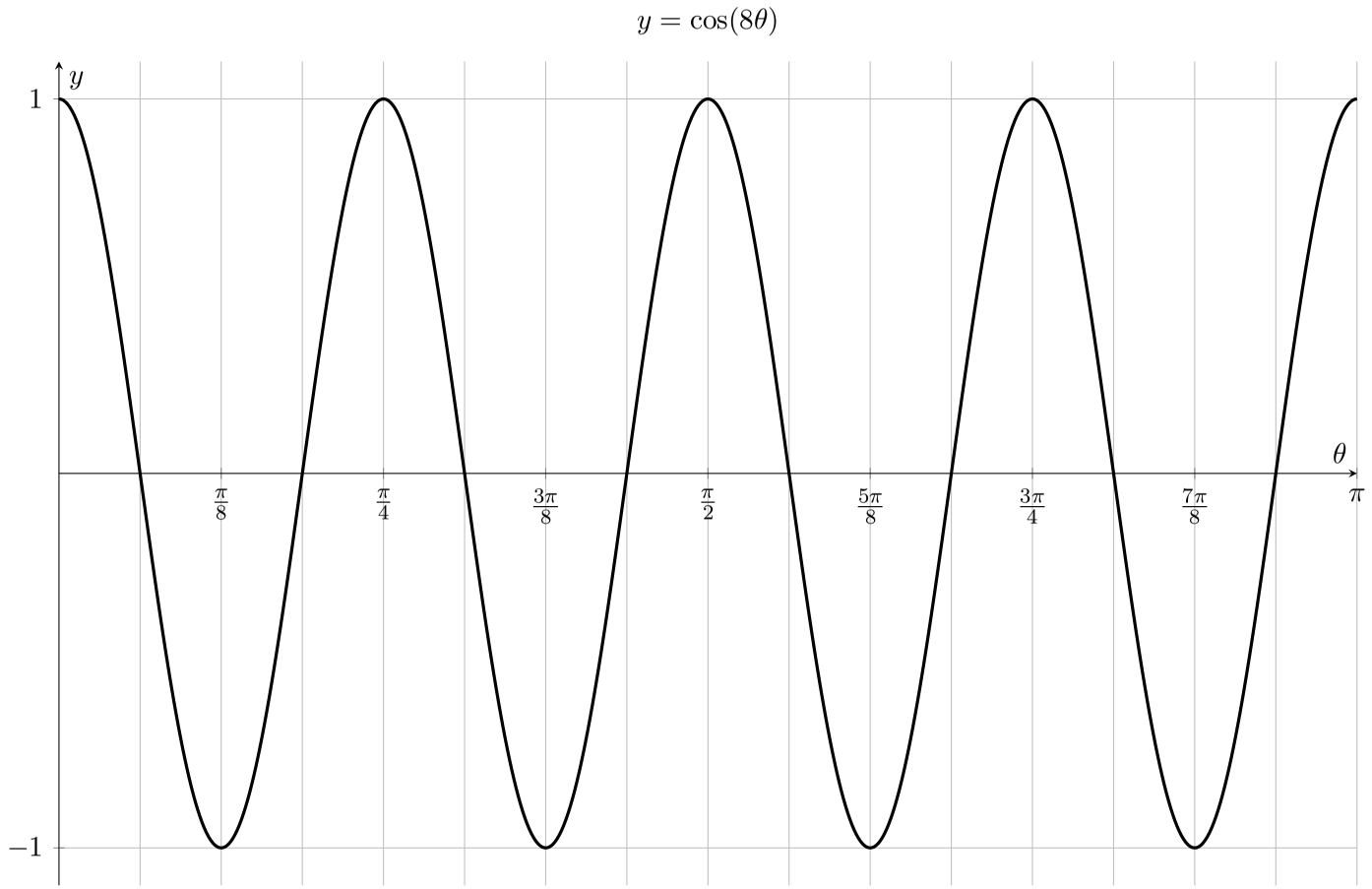
is moved to

$$(\cos\theta, \cos(n\theta)).$$

As θ runs from 0 to π, the new horizontal coordinate $x = \cos\theta$ runs from 1 to -1. The resulting curve is precisely the graph of the Chebyshev polynomial, since

$$T_n(\cos\theta) = \cos(n\theta).$$

In this sense, a Chebyshev polynomial is simply a cosine wave viewed through the nonlinear change of horizontal coordinate $x = \cos\theta$. The equally spaced oscillations in θ become compressed near $x = -1$ and $x = 1$.


$y = \cos(8\theta)$
y
1
-1
θ
$\frac{\pi}{8}$
$\frac{\pi}{4}$
$\frac{3\pi}{8}$
$\frac{\pi}{2}$
$\frac{5\pi}{8}$
$\frac{3\pi}{4}$
$\frac{7\pi}{8}$
π

Some properties of Chebyshev polynomials

Prove the following properties.

1. T_n is a polynomial of degree n.

2. For $n \geq 1$, the leading coefficient of T_n is 2^{n-1}. In other words,
$$T_n(x) = 2^{n-1}x^n + \text{terms of lower degree.}$$

3. For every $x \in [-1, 1]$,
$$|T_n(x)| \leq 1.$$

4. Show that
$$T_n(-x) = (-1)^n T_n(x).$$
Thus, T_n is even when n is even and odd when n is odd.

5. Determine the values of $T_n(1)$ and $T_n(-1)$. Determine the values of $T_n(0)$.

6. Show that the sum of the coefficients of $T_n(x)$ is 1.

   answer 1: because $T_n(1) = 1$ and because for any polynomial the value at 1 is the sum of the coefficients, we have the results.
   answer 2: we can also prove this by induction.

Zeros of the Chebyshev polynomials

The polynomial T_n has exactly n zeros in the interval $[-1, 1]$. They are given by

$$\boxed{x_k = \cos\left(\frac{(2k-1)\pi}{2n}\right), \qquad k = 1, \ldots, n.}$$

Proof. For $x \in [-1, 1]$, write

$$x = \cos\theta, \qquad 0 \le \theta \le \pi.$$

By the definition of the Chebyshev polynomial,

$$T_n(x) = T_n(\cos\theta) = \cos(n\theta).$$

Therefore,

$$T_n(x) = 0 \quad \Longleftrightarrow \quad \cos(n\theta) = 0.$$

The zeros of the cosine function occur at

$$n\theta = \frac{(2k-1)\pi}{2}, \qquad k \in \mathbb{Z}.$$

Restricting to $0 \le \theta \le \pi$ gives

$$\theta_k = \frac{(2k-1)\pi}{2n}, \qquad k = 1, \ldots, n.$$

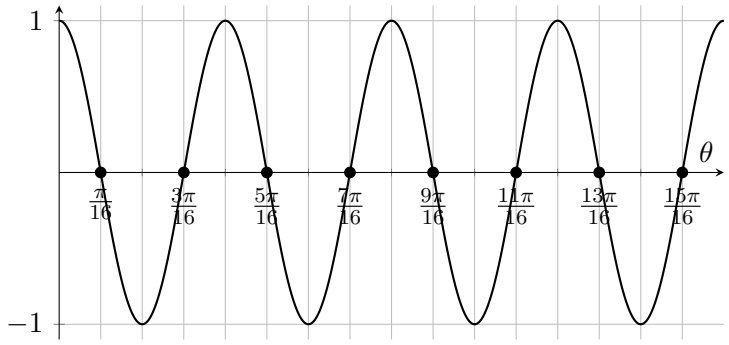
Hence

$$x_k = \cos\theta_k = \cos\left(\frac{(2k-1)\pi}{2n}\right), \qquad k = 1, \ldots, n.$$

These values are distinct and belong to $(-1, 1)$, so they give the n zeros of the degree-n polynomial T_n. □

The zeros of T_8

$y = \cos(8\theta)$ for $0 \le \theta \le \pi$



For $n = 8$, the zeros are

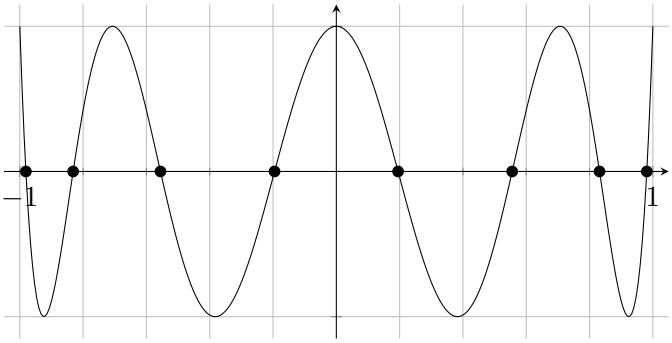
$$x_k = \cos\left(\frac{(2k-1)\pi}{16}\right), \qquad k = 1, \ldots, 8.$$

The relation between these zeros and the angles

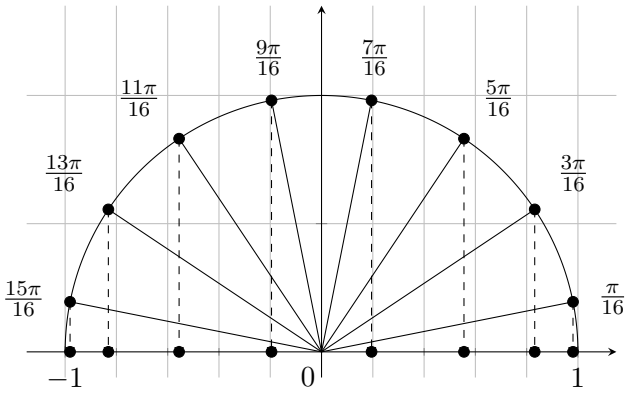
$$\theta_k = \frac{(2k-1)\pi}{16}$$

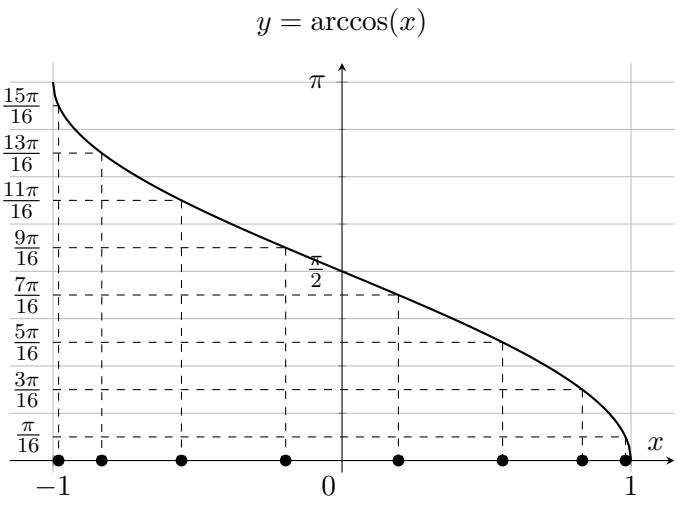
can be visualized in three ways.

$$T_8(x) = 128x^8 - 256x^6 + 160x^4 - 32x^2 + 1$$



The zeros of T_8 on the unit circle





The values $+1$ and -1

For $x \in [-1, 1]$, write $x = \cos\theta$, where $0 \le \theta \le \pi$. Since

$$T_n(x) = \cos(n\theta),$$

we have

$$T_n(x) = 1 \quad \Longleftrightarrow \quad n\theta = 2k\pi,$$

and

$$T_n(x) = -1 \quad \Longleftrightarrow \quad n\theta = (2k+1)\pi.$$

Thus, the points where T_n takes the values $+1$ and -1 are

$$\boxed{T_n(x) = 1 \quad \Longleftrightarrow \quad x = \cos\left(\frac{2k\pi}{n}\right)}$$

and

$$\boxed{T_n(x) = -1 \quad \Longleftrightarrow \quad x = \cos\left(\frac{(2k+1)\pi}{n}\right),}$$

where k is restricted so that the corresponding angle belongs to $[0, \pi]$.

Equivalently, for $k = 0, \ldots, n$,

$$\boxed{T_n\left(\cos\frac{k\pi}{n}\right) = (-1)^k.}$$

These $n+1$ points are the points at which T_n alternates between $+1$ and -1.

If n is even, the alternation starts and ends at $+1$. Thus, among these $n+1$ points, T_n takes the value $+1$ at $\frac{n}{2}+1$ points and the value -1 at $\frac{n}{2}$ points.

If n is odd, the alternation starts at $+1$ and ends at -1. Thus, T_n takes each of the values $+1$ and -1 at $\frac{n+1}{2}$ points.

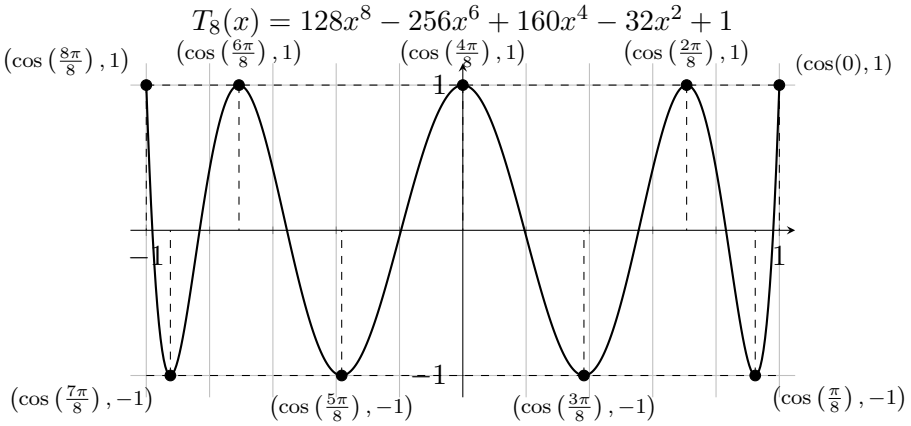
Example: T_8

For $n = 8$,

$$T_8\left(\cos\frac{k\pi}{8}\right) = (-1)^k, \qquad k = 0, \ldots, 8.$$

Thus,

$$\begin{aligned} T_8(1) &= 1, \\ T_8\left(\cos\frac{\pi}{8}\right) &= -1, \\ T_8\left(\cos\frac{2\pi}{8}\right) &= 1, \\ T_8\left(\cos\frac{3\pi}{8}\right) &= -1, \\ &\vdots \\ T_8\left(\cos\pi\right) &= 1. \end{aligned}$$



The minimax property

Theorem. Among all monic polynomials of degree n, the polynomial

$$\frac{1}{2^{n-1}} T_n(x)$$

minimizes the maximum absolute value on $[-1, 1]$. In other words,

$$\boxed{\min_{\substack{p \text{ monic} \\ \deg(p)=n}} \max_{-1 \le x \le 1} |p(x)| = \frac{1}{2^{n-1}}.}$$

The minimum is attained by

$$\boxed{p(x) = \frac{1}{2^{n-1}} T_n(x).}$$

Proof. Let p be any monic polynomial of degree n and define

$$q(x) = p(x) - \frac{1}{2^{n-1}} T_n(x).$$

Since p is monic and the leading coefficient of T_n is 2^{n-1}, the polynomial $\frac{1}{2^{n-1}} T_n$ is also monic. Hence the leading terms cancel and

$$\deg q \le n - 1.$$

Now consider the $n + 1$ points

$$x_k = \cos\left(\frac{k\pi}{n}\right), \qquad k = 0, \ldots, n.$$

We have

$$T_n(x_k) = T_n\left(\cos\frac{k\pi}{n}\right) = \cos(k\pi) = (-1)^k.$$

Suppose, for contradiction, that

$$\max_{-1 \le x \le 1} |p(x)| < \frac{1}{2^{n-1}}.$$

Then

$$|p(x_k)| < \frac{1}{2^{n-1}},$$

and therefore

$$q(x_k) = p(x_k) - \frac{1}{2^{n-1}}(-1)^k$$

has the sign of $-(-1)^k$. Thus the values

$$q(x_0), q(x_1), \ldots, q(x_n)$$

alternate in sign.

Since the points x_k are distinct, the intermediate value theorem implies that q has at least one zero between each two consecutive points. Hence q has at least n distinct zeros. This is impossible because $\deg q \leq n-1$, unless q is the zero polynomial.

Therefore,

$$\max_{-1\leq x\leq 1} |p(x)| \geq \frac{1}{2^{n-1}}.$$

On the other hand, since

$$|T_n(x)| \leq 1, \qquad -1 \leq x \leq 1,$$

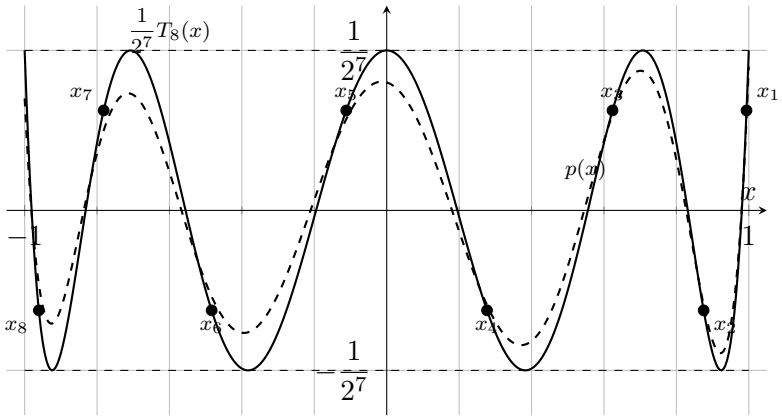
we have

$$\max_{-1\leq x\leq 1} \left|\frac{1}{2^{n-1}} T_n(x)\right| = \frac{1}{2^{n-1}}.$$

Thus the minimum is attained by $\frac{1}{2^{n-1}}T_n$.

□

$\frac{1}{2^7}T_8(x)$ and $p(x)$ with $\max_{[-1,1]} |p(x)| < \frac{1}{2^7}$



Chebyshev nodes and polynomial interpolation

We first recall the interpolation error formula.

Theorem (Interpolation error). Let $x_1, \dots, x_n$ be distinct points in $[a, b]$, and let p_{n-1} be the polynomial of degree at most $n-1$ that interpolates f at these points. If f is n times continuously differentiable on $[a, b]$, then, for every $x \in [a, b]$, there exists a point $\xi = \xi(x) \in (a, b)$ such that

$$\boxed{f(x) - p_{n-1}(x) = \frac{f^{(n)}(\xi)}{n!} \prod_{j=1}^{n} (x - x_j).}$$

Consequently,

$$|f(x) - p_{n-1}(x)| \leq \frac{\max_{\xi \in [a,b]} |f^{(n)}(\xi)|}{n!} \left| \prod_{j=1}^{n} (x - x_j) \right|.$$

Taking the maximum over $x \in [a, b]$ gives

$$\boxed{\max_{x \in [a,b]} |f(x) - p_{n-1}(x)| \leq \frac{\max_{\xi \in [a,b]} |f^{(n)}(\xi)|}{n!} \max_{x \in [a,b]} \left| \prod_{j=1}^{n} (x - x_j) \right|.}$$

We now specialize to the interval $[-1, 1]$. The first factor in this bound depends only on the function f. The choice of interpolation nodes affects the second factor. It is therefore natural to ask how the nodes $x_1, \dots, x_n$ should be chosen so that

$$\max_{-1 \leq x \leq 1} |(x - x_1)(x - x_2) \cdots (x - x_n)|$$

is as small as possible.

Theorem. Among all choices of n nodes

$$x_1, \dots, x_n \in [-1, 1],$$

the quantity

$$\max_{-1 \leq x \leq 1} |(x - x_1)(x - x_2) \cdots (x - x_n)|$$

is minimized when the nodes are chosen as the zeros of T_n.

In other words,

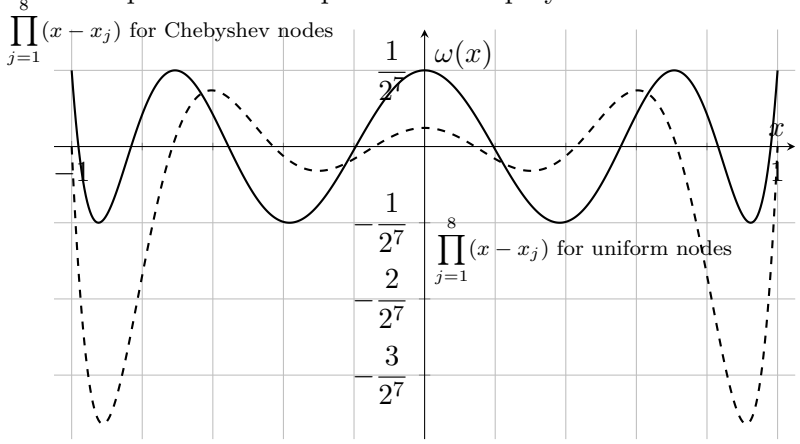
$$\boxed{\min_{x_1,\ldots,x_n\in[-1,1]} \max_{-1\le x\le 1} \left|\prod_{j=1}^{n}(x-x_j)\right| = \frac{1}{2^{n-1}}.}$$

The minimum is attained for the Chebyshev nodes

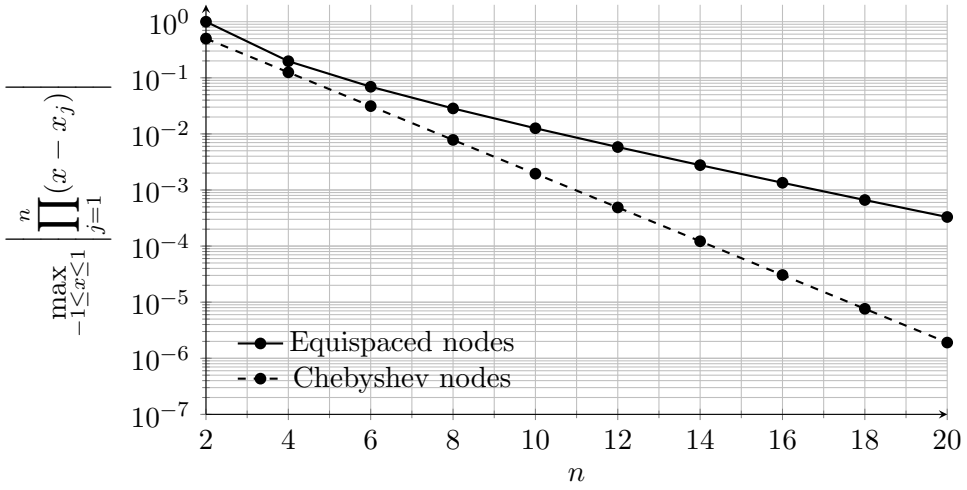
$$\boxed{x_j = \cos\left(\frac{(2j-1)\pi}{2n}\right), \qquad j=1,\ldots,n.}$$

Equispaced nodes versus Chebyshev nodes

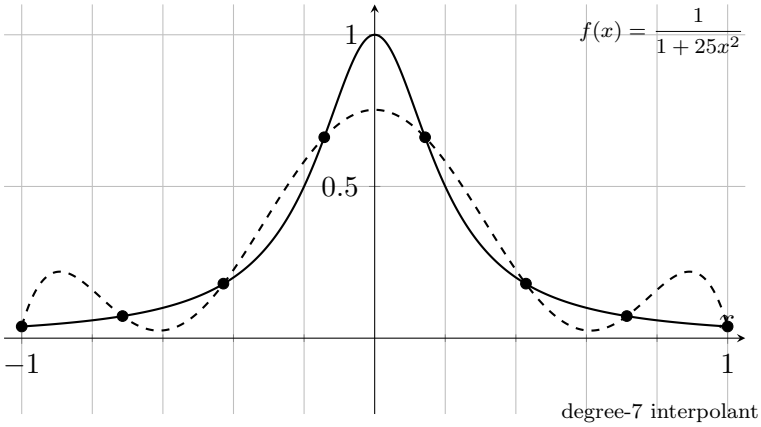
Comparison of interpolation node polynomials for $n = 8$



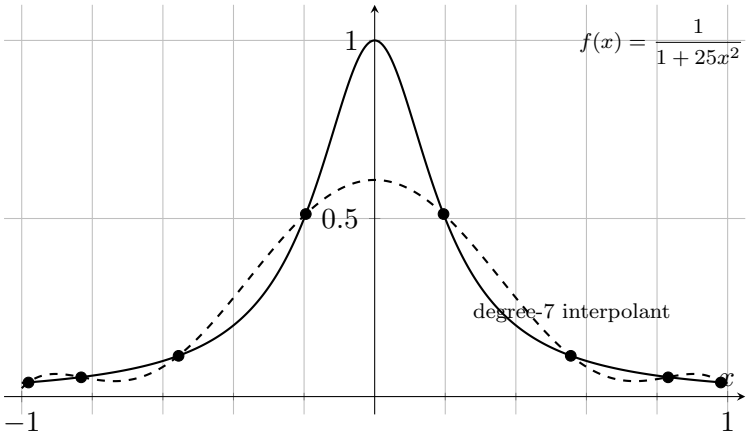
Equispaced nodes versus Chebyshev nodes



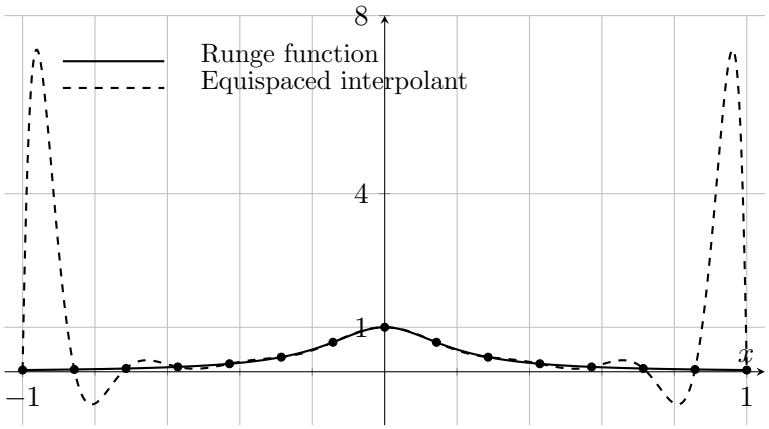
Runge function and degree-7 interpolation with 8 equispaced nodes



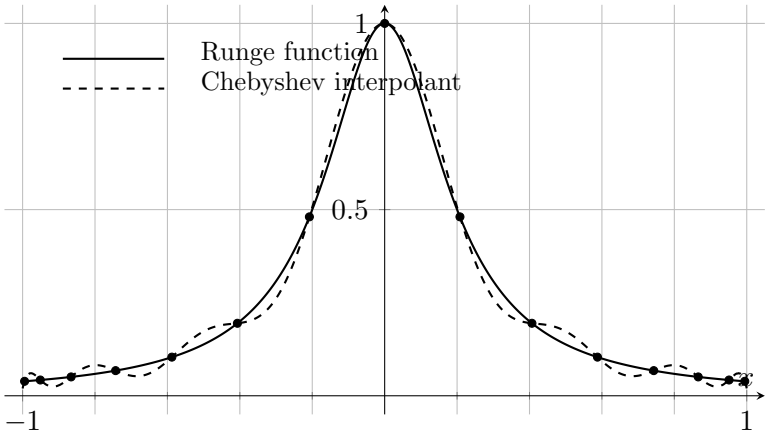
Runge function and degree-7 interpolation with 8 Chebyshev nodes



Runge function and degree-14 interpolation with 15 equispaced nodes



Runge function and degree-14 interpolation with 15 Chebyshev nodes



Chebyshev nodes on a general interval

So far, we have considered interpolation on the interval $[-1, 1]$. The Chebyshev nodes can easily be mapped to any interval $[a, b]$.

The affine transformation

$$x = \frac{a+b}{2} + \frac{b-a}{2}t$$

maps $t \in [-1, 1]$ onto $x \in [a, b]$.

Therefore, the n Chebyshev nodes on $[a, b]$ are

$$\boxed{x_j = \frac{a+b}{2} + \frac{b-a}{2}\cos\left(\frac{(2j-1)\pi}{2n}\right), \qquad j = 1, \ldots, n.}$$

Thus, the Chebyshev nodes on $[a, b]$ are obtained simply by scaling and translating the Chebyshev nodes on $[-1, 1]$.

The minimax value is also scaled. Indeed, let

$$t = \frac{2x - a - b}{b - a}.$$

Then

$$x - x_j = \frac{b-a}{2}(t - t_j),$$

where

$$t_j = \cos\left(\frac{(2j-1)\pi}{2n}\right).$$

Consequently,

$$\prod_{j=1}^{n}(x-x_j)=\left(\frac{b-a}{2}\right)^n\prod_{j=1}^{n}(t-t_j).$$

Since

$$\max_{-1\le t\le 1}\left|\prod_{j=1}^{n}(t-t_j)\right|=\frac{1}{2^{n-1}},$$

we obtain

$$\boxed{\min_{x_1,\ldots,x_n\in[a,b]}\max_{a\le x\le b}\left|\prod_{j=1}^{n}(x-x_j)\right|=\left(\frac{b-a}{2}\right)^n\frac{1}{2^{n-1}}=\frac{(b-a)^n}{2^{2n-1}}.}$$

Equivalently, the optimal monic polynomial on $[a,b]$ is

$$\boxed{\left(\frac{b-a}{2}\right)^n\frac{1}{2^{n-1}}T_n\left(\frac{2x-a-b}{b-a}\right).}$$

Example: eight nodes on $[2,7]$

Consider $n=8$ interpolation nodes on the interval $[2,7]$. Here,

$$\frac{a+b}{2}=\frac{9}{2},\qquad \frac{b-a}{2}=\frac{5}{2}.$$

Therefore, the eight Chebyshev nodes are

$$\boxed{x_j=\frac{9}{2}+\frac{5}{2}\cos\left(\frac{(2j-1)\pi}{16}\right),\qquad j=1,\ldots,8.}$$

These are obtained by taking the eight zeros of T_8 on $[-1,1]$, stretching them by a factor of $5/2$, and translating them by $9/2$.

The corresponding minimax value is

$$\begin{aligned}\min_{x_1,\ldots,x_8\in[2,7]}\max_{2\le x\le 7}\left|\prod_{j=1}^{8}(x-x_j)\right|&=\left(\frac{5}{2}\right)^8\frac{1}{2^7}\\&=\frac{5^8}{2^{15}}.\end{aligned}$$